Anthropogenic forcing of behavioral plasticity and consequences for ecosystem structure

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Abstract

The role of group personality is crucial in understanding the dynamics of predator-prey systems. Our study enhances the Rosenzweig-MacArthur model by incorporating personality through a Holling type-II boldness function to modify the attack rate and adding a boldness-dependent harvesting term to account for external harvesting costs on predators. Mathematically, we focus on analyzing the stability of coexisting equilibrium points and identifying all potential bifurcations within the system. We illustrate the system's behavior under various parameter settings through bifurcation diagrams and phase portraits. Our findings reveal the presence of transcritical bifurcation, saddle-node bifurcation, and Hopf bifurcation points, providing deeper insights into the complex dynamics of predator-prey interactions.

1 Introduction

Studying predator-prey models is essential for understanding the dynamics of ecosystems and the interactions between species, which is crucial for conservation efforts and biodiversity management. Additionally, these models help predict the impacts of environmental changes and inform policy decisions, making them vital for scientific research and practical applications in natural resource management and conservation planning.

Several predator-prey models attempt to capture these dynamics, such as the Lotka-Volterra Model [15] and the Rosenzweig-MacArthur Model [13], among others. However, these models often incorporate relatively simplistic interactions between species. Earlier models predominantly relied on group size to predict the population dynamics of the species involved. Recent studies, however, have demonstrated that additional factors play significant roles in these dynamics.

Keiser et al. [12] illustrated that a colony of ten bold spiders (Stegodyphus dumicola) executed as many attacks as a colony of 110 'average' spiders, indicating that the personality composition of the group is a critical factor in developing predator-prey models. Furthermore, studies have shown that bold individuals prioritize reproduction over self-maintenance, suggesting that personality influences the long-term population dynamics of species as well [4]. Although an individual's personality might seem abstract, recent research indicates that prey availability significantly affects the group's personality. Clermont et al. [2] proposed that bolder predators tend to reproduce more during periods of low prey availability. However, boldness has negative consequences as well. Studies have shown that bold individuals of a species are more likely to be harvested by external predators [10, 7, 1, 14, 17].

Keeping these considerations in mind, we aim to bridge this theoretical gap in our model. We modify the Rosenzweig and MacArthur model by incorporating a Holling type-II boldness function to the attack rate and adding a harvesting term proportional to boldness to the predator to represent the external harvesting cost on the predator. In the subsequent sections, we discuss the model formulation along with the positivity and boundedness of the solutions. The stability and existence of equilibria are also studied. Finally, we examine the effect of varying the model parameters in the bifurcation analysis of the model and discuss their implications.

2 The Model

The Rosenzweig and MacArthur (RM) model [13] is given by,

$$\begin{cases} \dfrac{dx}{dt} = rx\left(1 - \dfrac{x}{k}\right) - \dfrac{\beta xy}{1+\beta hx}, \\ \dfrac{dy}{dt} = e\dfrac{\beta xy}{1+\beta hx} - my \end{cases}$$

where x and y represent the population densities of prey and predators, respectively, both with positive initial conditions. The parameters are defined as follows: m denotes the natural death rate of predators, e signifies the conversion efficiency, h represents the handling time of the predator, r is the intrinsic growth rate of the prey, and k indicates the carrying capacity of the prey population. The term β denotes the attack rate.

We enhance the Rosenzweig-MacArthur model by integrating personality through a boldness functional response, influencing both the attack rate and predator mortality. In this revised model, predator mortality now includes an additional harvesting term proportional to predator boldness, and the attack rate is similarly adjusted to be proportional to boldness. Recognizing that low prey availability increases predator boldness, we implement a Holling Type-II functional response to accurately reflect this relationship in the modified model. The proposed model is now given by,

$$\begin{cases} \dfrac{dx}{dt} = rx\left(1 - \dfrac{x}{k}\right) - \dfrac{\beta(x,y)xy}{1+\beta(x,y)hx}, \\ \dfrac{dy}{dt} = e\dfrac{\beta(x,y)xy}{1+\beta(x,y)hx} - my - H(x,y)y \end{cases} \tag{1}$$

where,

$$\beta(x,y) = b_0 + \frac{a}{c\left(\dfrac{x}{y}\right)+1} = b_0 + \frac{ay}{cx+y}$$

$$H(x,y) = b_1\beta(x,y)$$

$$\begin{cases} \dfrac{dx}{dt} = rx\left(1 - \dfrac{x}{k}\right) - \dfrac{\beta(x,y)xy}{1+\beta(x,y)hx}, \\ \dfrac{dy}{dt} = e\dfrac{\beta(x,y)xy}{1+\beta(x,y)hx} - my - H(x,y)y \end{cases} \tag{2}$$

where,

$$\beta(x,y) = b_0 B(x,y)$$
$$H(x,y) = b_1 B(x,y)$$
$$B(x,y) = 1 + \frac{a}{c\left(\dfrac{x}{y}\right)+1} = 1 + \frac{ay}{cx+y}$$

We define b_0 as the minimum attack rate of the predator, a as the maximum increment of the attack rate, and c as the half-saturation constant for the boldness-based increment of the attack rate. Additionally, the mortality rate due to harvesting is represented by the functional response $H(x,y)$, with b_1 denoting the harvesting efficacy.

3 Model Analysis

3.1 Positivity and Boundedness of Solutions

The system of equations 2 is continuously differentiable in a neighborhood of the first quadrant $Q = \{(x,y) : x, y \geq 0\}$. As a result, solutions to initial value problems with non-negative initial conditions exist and are unique. In this section, we will demonstrate that Q is invariant, meaning that solutions starting with $x(0) \geq 0$ and $y(0) \geq 0$ will satisfy $x(t) \geq 0$ and $y(t) \geq 0$ for all t. Additionally, we will prove that these solutions are bounded in the future and thus are defined for all $t \geq 0$.

One can see that the system of equations 2 is of the form,

$$\frac{dx}{dt} = x' = xM(x,y)$$
$$\frac{dy}{dt} = y' = yN(x,y)$$

Clearly if $(x(t), y(t))$ is a solution of the system, then $x' = xm(t)$ and $y' = yn(t)$ where $m(t) = M(x(t), y(t))$ and $n(t) = N(x(t), y(t))$. Hence, upon using the integration factor trick, we get $\frac{d}{dt}(e^{-\int_0^t m(s)\,ds}x(t)) = \frac{d}{dt}(e^{-\int_0^t n(s)\,ds}y(t)) = 0$, which implies,

$$\begin{cases} x(t) = e^{\int_0^t m(t)\,dt}x(0) \\ y(t) = e^{\int_0^t n(t)\,dt}y(0) \end{cases} \tag{3}$$

Hence from the set of equations 3, we can see that if $(x(0), y(0)) \in Q$ then $(x(t), y(t)) \in Q$, thus proving invariance in Q.

For Boundedness, we prove the following:

Lemma 1. *There exists a positive constant S_0 such that for all $S \geq S_0$, the right triangle $T(S)$ with sides $x = 0$, $y = 0$, and $x + y = S$ is positively invariant.*

Since every initial point $(x(0), y(0)) \in Q$ also lies within $T(S)$ for some $S \geq S_0$, it follows that all solutions starting in Q are bounded for $t \geq 0$ because $(x(t), y(t)) \in T(S)$ for all $t \geq 0$.

Proof. From the previous section, we have seen that $x(t) \geq 0$ and $y(t) \geq 0$; hence, it suffices to prove that the solutions do not cross the hypotenuse given by $x + y = S$. For this, we show that for a solution (x, y) if $x + y = S$ then $\frac{d}{dt}(x + y) = x' + y' < 0$.

From our model,

$$x' + y' = rx\left(1 - \frac{x}{k}\right) - (1-e)\frac{\beta xy}{1+\beta hx} - my - H(x,y)y$$

From the positivity property of our solutions, we can see that,

$$x' + y' \leq rx\left(1 - \frac{x}{k}\right) - my = rx\left(1 - \frac{x}{k}\right) - m(S - x) = F(x)$$

$F(x)$ achieves it's maximum value at $\bar{x} = \frac{k}{2}(1 + m)$. Hence,

$$x' + y' \leq F(x) \leq -mS + \frac{k}{2}(1+m)^2 - \frac{1}{k}(\frac{k}{2}(1+m))^2 < 0$$

if we choose a large enough S. $\square$

3.2 Equilibria and their Stability

The equilibrium points of the system 2 are the points of intersection of the nullclines given by

$$\begin{cases} f_1(x,y) = rx\left(1 - \frac{x}{k}\right) - \frac{\beta(x,y)xy}{1+\beta(x,y)hx} = 0, \\ f_2(x,y) = e\frac{\beta(x,y)xy}{1+\beta(x,y)hx} - my - H(x,y)y = 0 \end{cases} \tag{4}$$

From the system 4, we see that there always exists the axial equilibrium $E_1 = (k, 0)$ along with the possibility of an interior equilibrium which are the points of intersection of the non-trivial nullclines 5 in the first quadrant.

$$\begin{cases} er(1 - \frac{x}{k}) = \frac{eb_0 y(cx+y+ay)}{cx(1+b_0hx)+y+(1+a)b_0hxy}, \\ \frac{eb_0 x(cx+y+ay)}{cx(1+b_0hx)+y+(1+a)b_0hxy} = m + b_1(1 + \frac{ay}{cx+y}) \end{cases} \tag{5}$$

In the first quadrant, the prey nullcline is a monotonically decreasing curve connecting the points $(0, \frac{r}{(a+1)b_0})$ and the axial equilibrium $(k, 0)$. Conversely, the predator nullcline is a monotonically non-decreasing curve that intersects the x-axis at $(\frac{b_1+m}{b_0(-b_1h+e-hm)}, 0)$.

From Figure 1, it is evident that an increase in b_1 shifts the predator nullcline to the right. The system is capable of exhibiting up to two interior equilibria, denoted as $E_{1*}(x_{1*}, y_{1*})$ and $E_{2*}(x_{2*}, y_{2*})$, where $x_{1*} < x_{2*}$.

The Jacobian matrix of the system evaluated at $E(x, y)$ is given by

$$J(x,y) = \begin{bmatrix} r - \frac{2rx}{k} - \frac{b_0 y(t_1(x,y)+ay)}{t_4(x,y)} + \frac{t_6(x,y)}{(t_1(x,y)t_4(x,y))^2} + \frac{t_5(x,y)}{t_1(x,y)^2 t_2(x) t_3(x,y)} & -t_{10}(x,y) \\ \frac{y(t_7(x,y)+t_8(x,y)+t_9(x,y))}{(t_1(x,y)t_4(x,y))^2} & -t_{11}(x,y) + b_1\left(-1 - \frac{ay(2cx+y)}{t_1(x,y)^2}\right) \end{bmatrix} \tag{6}$$

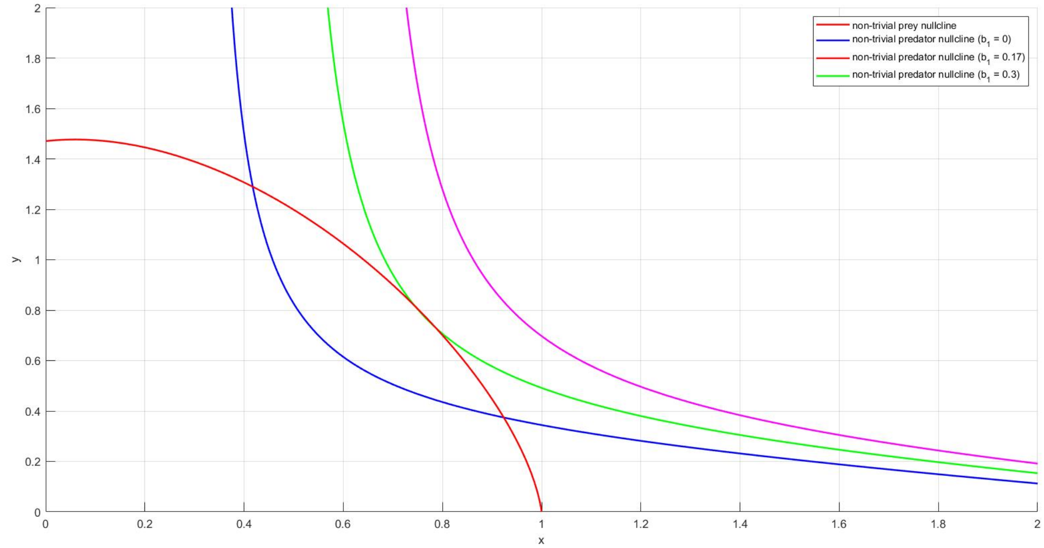


Figure 1: Non-trivial nullclines for $b_1 = 0$, 0.017, and 0.03 with the rest of the parameters having values from Table 1.

where,

$$
\begin{aligned}
t_1(x,y) &= cx + y \\
t_2(x) &= 1 + b_0 hx \\
t_3(x,y) &= 1 + \frac{ay}{t_1(x,y)} \\
t_4(x,y) &= cxt_2(x) + y + (1+a)b_0 hxy \\
t_5(x,y) &= ab_0 cxy^2 \\
t_6(x,y) &= b_0^2 hxy(cx + y + ay)(c^2x^2 + 2cxy + (1+a)y^2) \\
t_7(x,y) &= ab_1 cy(cx+y)^2 \\
t_8(x,y) &= ab_0^2 b_1 ch^2 x^2 y(cx + y + ay)^2 \\
t_9(x,y) &= b_0(cx+y)\left(c^3ex^3 + c^2(3e + 2ab_1 h)x^2 y\right. \\
&\quad + c\left((3+a)e + 2a(1+a)b_1 h\right)xy^2 \\
&\quad \left. + (1+a)ey^3\right)
\end{aligned}
$$

$$
t_{10}(x,y) = \frac{b_0 x\left(c^2x^2(1+b_0hx) + 2(1+a)cx(1+b_0hx)y + (1+a)\left(1 + (1+a)b_0hx\right)y^2\right)}{t_4(x,y)^2}
$$

$$
\begin{aligned}
t_{11}(x,y) &= \frac{c^2x^2(1+b_0hx)(m - b_0ex + b_0hmx) + 2cx(1+b_0hx)(m - (1+a)b_0ex + (1+a)b_0hmx)y}{t_4(x,y)^2} \\
&\quad + \frac{(1+(1+a)b_0hx)(m - (1+a)b_0ex + (1+a)b_0hmx)y^2}{t_4(x,y)^2}
\end{aligned}
$$

Proposition 2. *The following properties hold for the model 2:*

i The axial equilibrium E_1 is asymptotically stable if $m + b_1(1 + b_0hk) > b_0k(e - hm)$ and is a saddle point if $m + b_1(1 + b_0hk) < b_0k(e - hm)$.

ii The interior equilibrium $E_{1*}(x_{1*}, y_{1*})$ is asymptotically stable if the trace of the Jacobian, $\mathrm{Tr}(J(x_{1*}, y_{1*})) < 0$, and is unstable if $\mathrm{Tr}(J(x_{1*}, y_{1*})) > 0$. Meanwhile, $E_{2*}(x_{2*}, y_{2*})$ is always a saddle point.

Proof. i The Jacobian matrix at E_1 is:

$$J(E_1) = \begin{bmatrix} -1 & -\frac{b_0 k}{1+b_0 hk} \\ 0 & \frac{b_0 k(e-hm)-m}{1+b_0 hk} - b_1 \end{bmatrix}$$

The eigenvalues of this matrix are -1 and $\frac{b_0 k(e-hm)-m}{1+b_0 hk} - b_1$. Thus, E_1 is asymptotically stable if $m + b_1(1 + b_0 hk) > b_0 k(e - hm)$ and is a saddle point if $m + b_1(1 + b_0 hk) < b_0 k(e - hm)$.

ii Let $f_1(x, y) = xM(x, y)$ and $f_2(x, y) = yN(x, y)$. The Jacobian is then given by:

$$J(x, y) = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} x\frac{\partial M}{\partial x} & x\frac{\partial M}{\partial y} \\ y\frac{\partial N}{\partial x} & y\frac{\partial N}{\partial y} \end{bmatrix}$$

Let the gradients of the tangents for prey and predator nullclines at the interior equilibrium point be denoted by $\frac{dy^{(M)}}{dx}$ and $\frac{dy^{(N)}}{dx}$ respectively. By the implicit function theorem, we have:

$$\det(J(x, y)) = xy\frac{\partial M}{\partial y}\frac{\partial N}{\partial y}\left(\frac{dy^{(N)}}{dx} - \frac{dy^{(M)}}{dx}\right)$$

It can be seen that $\frac{\partial N}{\partial y} > 0$ and $\frac{\partial M}{\partial y} < 0$ in the first quadrant. From Figure 1, we also see that $\left.\frac{dy^{(N)}}{dx}\right|_{E_{1*}} < \left.\frac{dy^{(M)}}{dx}\right|_{E_{1*}}$ and $\left.\frac{dy^{(N)}}{dx}\right|_{E_{2*}} > \left.\frac{dy^{(M)}}{dx}\right|_{E_{2*}}$. Hence, $\det(J(x_{1*}, y_{1*})) > 0$ and $\det(J(x_{2*}, y_{2*})) < 0$.

This implies that E_{1*} is asymptotically stable if the trace of the Jacobian, $\mathrm{Tr}(J(x_{1*}, y_{1*})) < 0$, and is unstable if $\mathrm{Tr}(J(x_{1*}, y_{1*})) > 0$, while E_{2*} is always a saddle point. □

4 Results

We make use of the following values from Table 1 for the parameters to infer our results from the harvesting proportional to boldness model unless specified otherwise,

4.1 Variation in the System according to a Single Parameter

Figure 2 shows the bifurcation diagram for the system under different carrying capacities. Here, we can see that predators go extinct due to low food availability at lower carrying capacities. Upon further increasing the carrying capacity, there is a stable coexistence between the predator and prey population, with a sufficient food supply for the predators. At higher carrying capacities, there is an oscillating coexistence between the predator and prey population due to an excess food supply to the predators.

From Figure 3, which illustrates the variation in the system's state for different Minimum Attack Rates, we can observe that predators go extinct at lower Minimum Attack Rates due

Parameter	Values Used
Producer Carrying Capacity (k)	1
Maximal Growth Rate of Producer (r)	2.5
Natural Death Rate (m)	0.33
Maximal production efficiency (e)	0.7
Handling time (h)	0.25
Minimum attack rate of Predator (b_0)	0.2
Maximum increment in Boldness (a)	7.5
Half-saturation constant for Boldness (c)	1.2
Harvesting Efficacy (b_1)	0

Table 1: Parameter values used in developing results

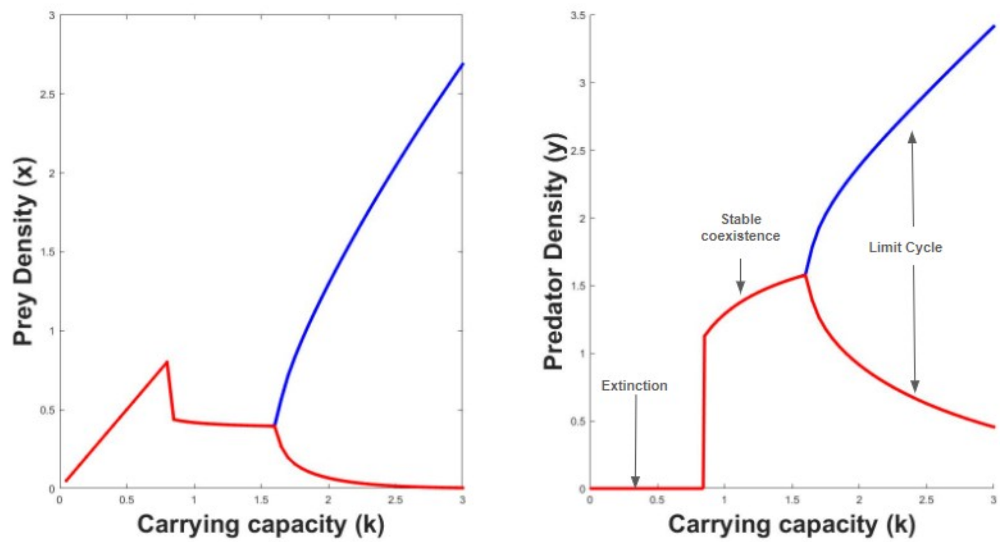


Figure 2: Bifurcation diagram of internal positive equilibrium (E^*) for the model under different carrying capacity (k).

to not hunting enough food. As the Minimum Attack Rate increases, a stable coexistence emerges between the predator and prey populations, with predators hunting adequate food. However, the populations exhibit oscillating coexistence at higher minimum attack rates because predators hunt for excess food.

The variation in the system's state for different Maximum Increments in Boldness is illustrated in Figure 4. It is evident that at lower values of a, predators face extinction due to inadequate hunting. As a increases, the predator and prey populations achieve stable coexistence, with predators securing enough food. At even higher values of a, the predator and prey populations experience oscillating coexistence, driven by an overabundance of food for the predators.

Figure 5 shows the bifurcation diagram for the system under different harvesting efficacies. We can see a stable coexistence of the prey and predator populations until a high Harvesting Efficiency drives the predators to extinction.

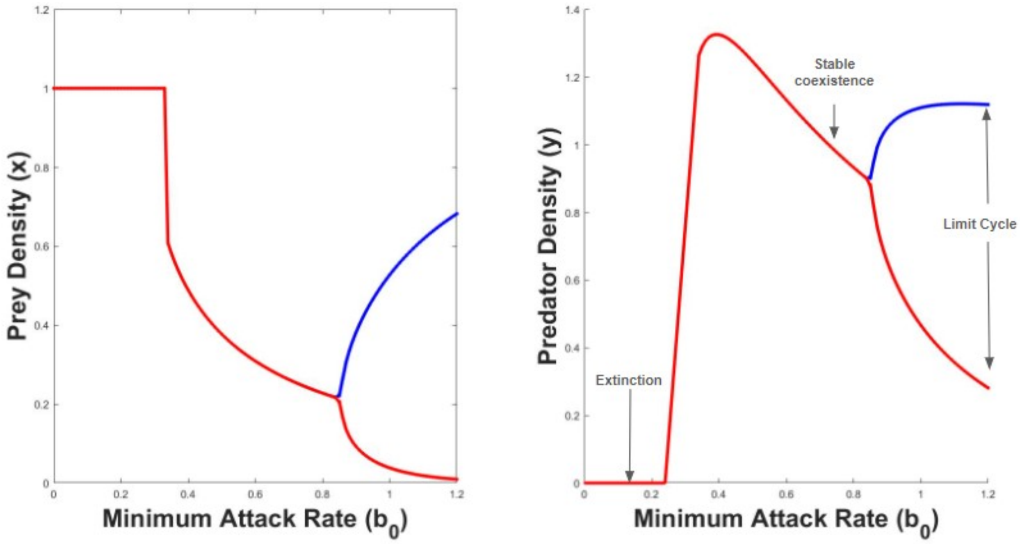


Figure 3: Bifurcation diagram of internal positive equilibrium (E^*) for the model under different Minimum Attack Rates (b_0) at $a = 2.5$.

4.2 Variation in the system according to Multiple Parameters

The whole bifurcation diagram from Figure 6a is divided into five sub-regions, labeled I through V, defined by the transcritical, saddle-node, and Hopf bifurcation curves. In region I (E), there are no interior equilibrium points. Moving from region I to III across the saddle-node bifurcation curve (red), two new interior equilibrium points, E_{1*} and E_{2*}, emerge, with E_{1*} being stable and E_{2*} being a saddle point. Transitioning from region III to II by crossing the transcritical bifurcation curve (black), one interior equilibrium point E_{2*} disappears, leaving E_{1*} stable while E_1 becomes unstable. Vertically moving from region III to V through the Hopf bifurcation curve (blue), E_{1*} loses stability, leading to the appearance of a stable limit cycle around E_{1*}, with E_{2*} remaining a saddle point. Finally, in the transition from region V to IV, the interior equilibrium point E_{2*} disappears, and E_1 becomes unstable.

To illustrate these transitions, we plot the phase portraits for each region in Figure 7, where blue curves represent trajectories, separatrix is represented by a red curve, the red solid circle represents a stable equilibrium point, and an open red circle represents an unstable equilibrium point while the pink dotted curve denotes the prey nullcline and the orange dotted curve denotes the predator nullcline.

Comparing Figure 6a with Figures 6b and 6c, it becomes evident that at lower prey carrying capacities (k) the risk of predator extinction is increased, with the Extinction Region becoming more dominant compared to regions of coexistence. We also see that at low carrying capacities, it is impossible to reach states of pure coexistence by varying other parameters of the system. This underscores the necessity of maintaining adequate prey populations to ensure the stability and sustainability of predator populations.

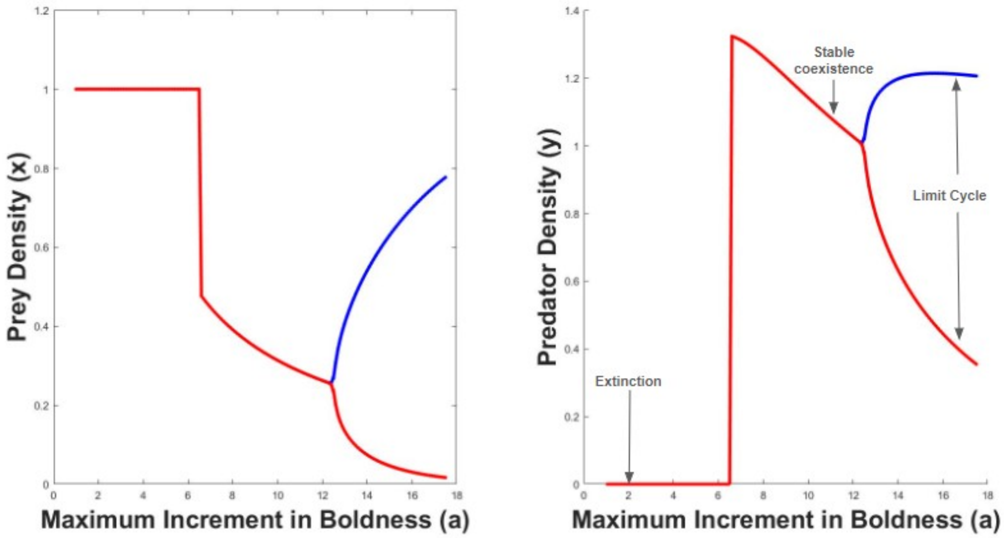


Figure 4: Bifurcation diagram of internal positive equilibrium (E^*) for the model under different Maximum Increments in Boldness (a).

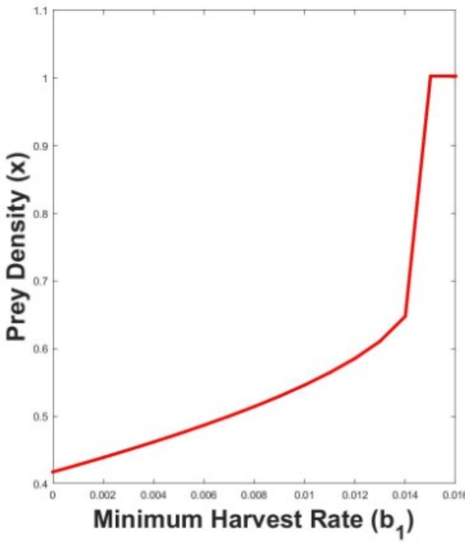
5 Discussion

Our study uncovers complex dynamics in predator-prey interactions, driven by variations in carrying capacity, minimum attack rate, predator boldness, and harvesting efficacy. These findings have important biological implications for understanding and managing ecosystems.

At low carrying capacities, there is a heightened risk of predator extinction due to insufficient prey availability, underscoring the importance of ensuring that the environment supports adequate prey populations [18, 6, 19]. As the carrying capacity increases, stable coexistence between predators and prey becomes possible [3]. However, at very high carrying capacities, where prey resources are abundant, the system often exhibits oscillatory behavior. These oscillations arise from cycles of prey over-exploitation, followed by depletion, and subsequent crashes in predator populations. This pattern is observed in the snowshoe hare-lynx system [8], where hares never face a food shortage, yet the population oscillates primarily due to predation pressure from lynxes. When carrying capacity falls below a critical threshold, predator extinction becomes inevitable, regardless of other parameters, emphasizing the need to carefully manage prey populations to maintain ecosystem stability.

The minimum attack rate is also crucial for predator survival. When attack rates are too low, predators cannot capture sufficient prey, leading to their extinction [16]. Effective hunting strategies, therefore, are essential for predator persistence. As attack rates increase, stable coexistence is achieved, as seen in ecosystems like the Isle Royale Wolf-Moose system [9], where moderate kill rates contribute to population stability. However, excessively high attack rates can destabilize the system, leading to oscillations. This highlights the need for predators to balance their hunting efforts to prevent prey depletion and avoid destabilizing their own populations, a phenomenon also observed in the hare-lynx dynamic [8].

Our findings on predator boldness show that low boldness results in insufficient prey capture, ultimately leading to predator extinction. As boldness increases, predators improve their hunting success and achieve stable coexistence with their prey. However, beyond a certain boldness threshold, the system enters a state of oscillation due to over-exploitation of prey. This



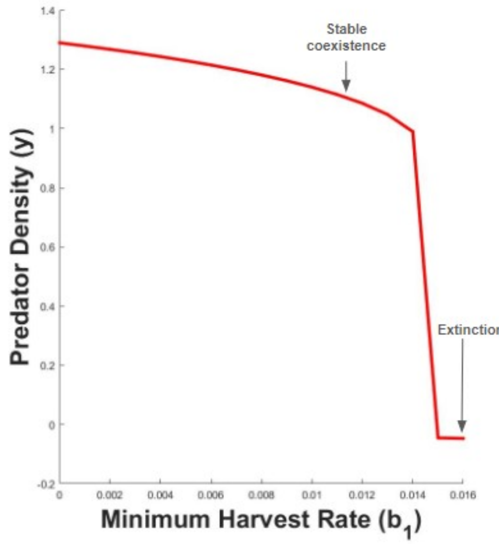
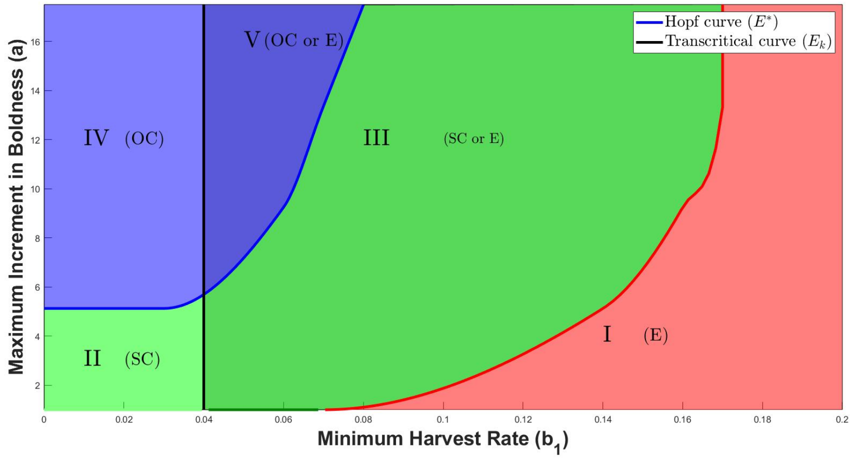


Figure 5: Bifurcation diagram of internal positive equilibrium (E^*) for the model under different Harvesting Efficacies (b_1)

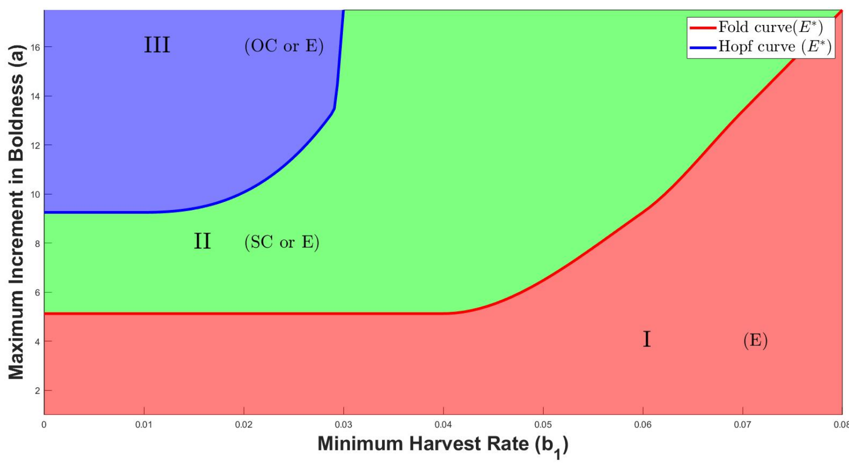
suggests that while boldness can enhance predator survival, it must be moderated to avoid destabilizing the ecosystem through excessive predation.

Harvesting efficacy plays a pivotal role in predator-prey dynamics as well. Moderate levels of harvesting promote stable coexistence, demonstrating that sustainable harvesting practices can help maintain ecosystem balance [5]. However, high harvesting efficiencies can drive predators to extinction, emphasizing the dangers of overharvesting [11, 5]. This highlights the importance of implementing sustainable management strategies that ensure the long-term survival of both predator and prey populations. The interplay between boldness and harvesting reveals complex dynamics, with regions of stable coexistence, oscillatory behavior, and extinction depending on the specific parameter values. This complexity suggests that effective ecosystem management must account for multiple interacting factors to maintain stability.

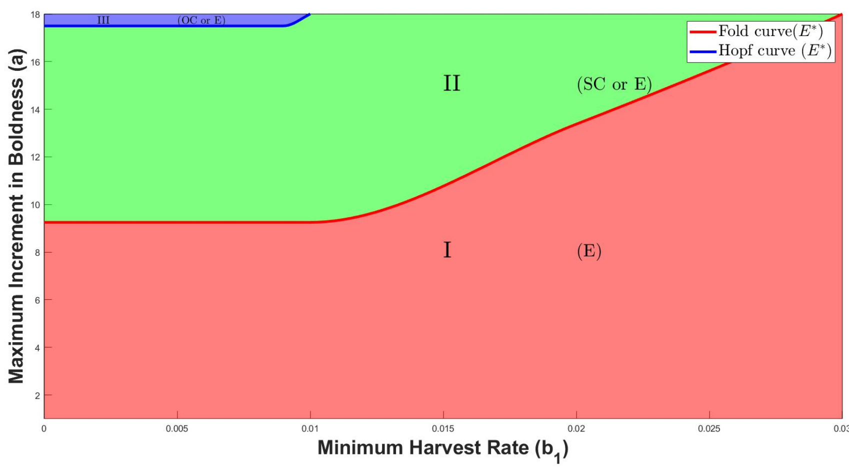
Overall, our study underscores the delicate balance required to sustain predator-prey ecosystems. Maintaining adequate prey populations, optimizing predator foraging strategies, and adopting sustainable harvesting practices are essential to prevent extinction and ensure long-term ecological stability. These findings highlight the need to consider both ecological interactions and behavioral adaptations when developing strategies for ecosystem management. Future research should focus on understanding adaptive responses to environmental changes and developing management practices that enhance the resilience of ecosystems.



(a) At High Carrying Capacity ($k = 3$).

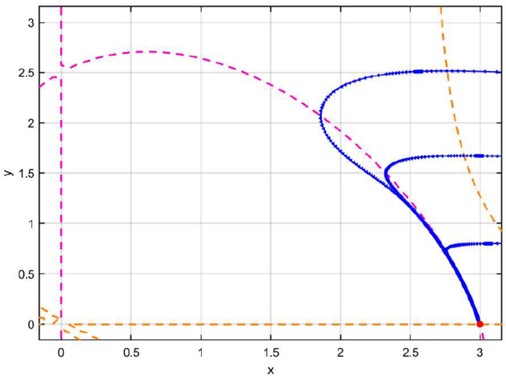


(b) At Medium Carrying Capacity ($k = 1.5$).

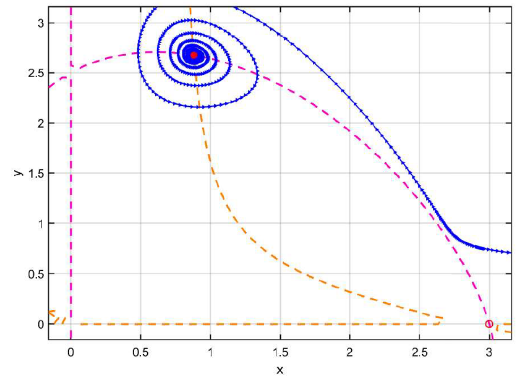


(c) At Low Carrying Capacity ($k = 0.75$).

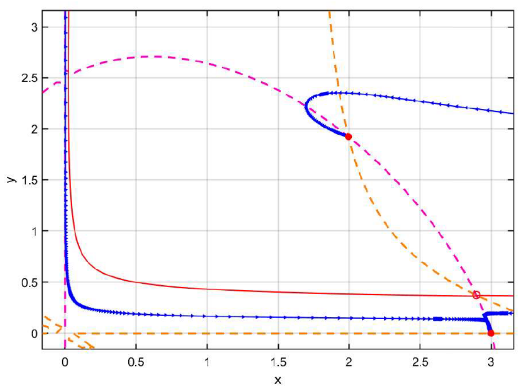
Figure 6: Two parameter bifurcation diagrams for different Maximum Increments in Boldness (a) and Harvesting Efficacies (b_1) at various carrying capacities.



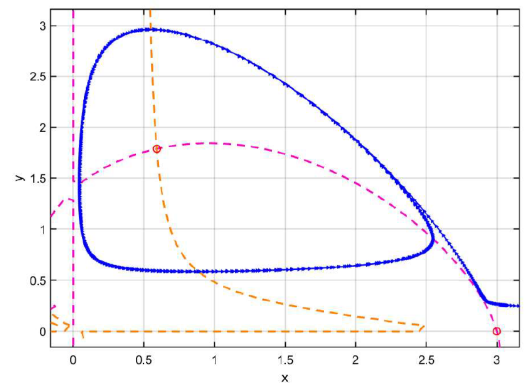
(a) At Extinction Region.



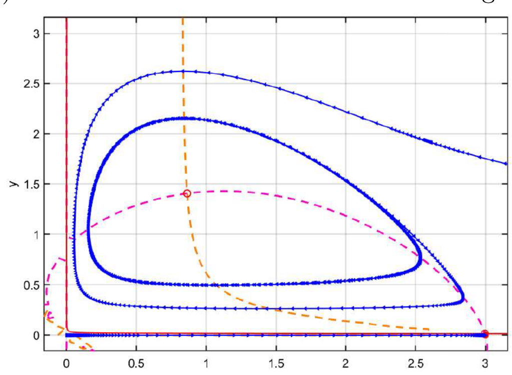
(b) At Stable Coexistence Region.



(c) At Stable Coexistence or Extinction Region.



(d) At Oscillating Coexistence Region



(e) At Oscillating Coexistence or Extinction Region

Figure 7: Phase portraits at different regions formed by varying a and b_1 for $k = 3$

References

[1] Robert Arlinghaus et al. “Passive gear-induced timidity syndrome in wild fish populations and its potential ecological and managerial implications”. In: *Fish and Fisheries* 18 (Mar. 2017), pp. 360–373. DOI: 10.1111/faf.12176.

[2] Lai S. et al. Clermont J. Couchoux C. “Prey availability influences the effect of boldness on reproductive success in a mammalian predator”. In: *Behav Ecol Sociobiol* 77 (2023).

[3] Ville-Petri Friman et al. “Availability of prey resources drives evolution of predator-prey interaction”. en. In: *Proc. Biol. Sci.* 275.1643 (July 2008), pp. 1625–1633.

[4] Stephanie M Harris et al. “Personality-specific carry-over effects on breeding”. In: *Proceedings of the Royal Society B: Biological Sciences* 287 (Dec. 2020). DOI: 10.1098/rspb.2020.2381.

[5] Ray Hilborn and Carl J. Walters. “Quantitative fisheries stock assessment: Choice, dynamics and uncertainty”. In: *Reviews in Fish Biology and Fisheries* 2 (1992), pp. 177–178. URL: https://doi.org/10.1007/BF00042883.

[6] Robert D Holt and Michael Barfield. “Trophic interactions and range limits: the diverse roles of predation”. en. In: *Proc. Biol. Sci.* 276.1661 (Apr. 2009), pp. 1435–1442.

[7] Kaj Hulthén et al. “A predation cost to bold fish in the wild”. In: *Scientific Reports* 7 (Dec. 2017). DOI: 10.1038/s41598-017-01270-w.

[8] Charles J. Krebs et al. “What Drives the 10-year Cycle of Snowshoe Hares?: The ten-year cycle of snowshoe hares—one of the most striking features of the boreal forest— is a product of the interaction between predation and food supplies, as large-scale experiments in the yukon have demonstrated”. In: *BioScience* 51.1 (Jan. 2001), pp. 25–35. ISSN: 0006-3568. DOI: 10.1641/0006-3568(2001)051[0025:WDTYCO]2.0.CO;2. eprint: https://academic.oup.com/bioscience/article-pdf/51/1/25/26890541/51-1-25.pdf. URL: https://doi.org/10.1641/0006-3568(2001)051[0025:WDTYCO]2.0.CO;2.

[9] L. David Mech. *The Wolves of Isle Royale*. University Press of the Pacific, 2002.

[10] Christopher Monk et al. “The battle between harvest and natural selection creates small and shy fish”. In: *Proceedings of the National Academy of Sciences* 118 (Mar. 2021), e2009451118. DOI: 10.1073/pnas.2009451118.

[11] Ransom A. Myers, Jeffrey A. Hutchings, and Nicholas J. Barrowman. “Why do Fish Stocks Collapse? The Example of Cod in Atlantic Canada”. In: *Ecological Applications* 7.1 (1997), pp. 91–106. ISSN: 10510761, 19395582. URL: http://www.jstor.org/stable/2269409 (visited on 09/02/2024).

[12] Keiser Carl N. and Pruitt Jonathan N. “Personality composition is more important than group size in determining collective foraging behaviour in the wild”. In: *Proc. R. Soc. B.* 281 (2014).

[13] M. L. Rosenzweig and R. H. MacArthur. “Graphical Representation and Stability Conditions of Predator-Prey Interactions”. In: *The American Naturalist* 97.895 (1963), pp. 209–223. ISSN: 00030147, 15375323. URL: http://www.jstor.org/stable/2458702 (visited on 08/28/2024).

[14] Judy Stamps. “Growth-mortality tradeoffs and “personality traits” in animals”. In: *Ecology letters* 10 (June 2007), pp. 355–63. DOI: 10.1111/j.1461-0248.2007.01034.x.

[15] V Volterra. "Variazioni e fluttuazioni del numero di individui in specie animali conviventi". In: *Atti Reale Accad. Nazionale dei Lincei* 6 (1927).

[16] John Vucetich et al. "Intra-seasonal variation in wolf Canis lupus kill rates". In: *Wildlife Biology* 18 (Sept. 2012), pp. 235–245. DOI: 10.2981/11-061.

[17] Ashley Ward et al. "Correlates of boldness in three-spined sticklebacks (Gasterosteus aculeatus)". In: *Behavioral Ecology and Sociobiology* 55 (Apr. 2004), pp. 561–568. DOI: 10.1007/s00265-003-0751-8.

[18] T C R White. "The role of food, weather and climate in limiting the abundance of animals". en. In: *Biol. Rev. Camb. Philos. Soc.* 83.3 (Aug. 2008), pp. 227–248.

[19] Christopher Wolf and William J Ripple. "Prey depletion as a threat to the world's large carnivores". en. In: *R. Soc. Open Sci.* 3.8 (Aug. 2016), p. 160252.